



Iborain Safety: The Community Public Safety Platform

Decentralized AI Transit Forensics & Crime Elimination Network • Nairobi, Kenya

Category: Entrepreneurship & Job Creation

Partner: ChipuRobo STEM Makerspaces

Target: Build with Gemini XPRIZE (\$2M)

1. EXECUTIVE SUMMARY & THE PROBLEM WE SOLVE

In Sub-Saharan Africa, over 70% of violent and property crime is transit-borne (getaway cars with cloned plates, unplated Boda Bodas). Flock Safety costs \$5,000/year (pricing out 99% of Africa), while dumb CCTV only records passive video without real-time deterrence. **Iborain Safety** deploys a \$49/mo infrastructure-free public safety grid powered by Google Gemini (Gemini 3.7 Flash & Live API) and locally fabricated edge sentries to eliminate crime in real time.

2. SENTRY HARDWARE SPECIFICATIONS (3 MODULAR TIERS)

TIER / MODEL	HARDWARE ARCHITECTURE	BOM COST	PRICING & MODEL	PAYBACK
Tier 1: Iborain Lite	Raspberry Pi Zero 2 W + 12MP Sony IMX708 + MAX98357A 3W Amp + MPU-6500 IMU	\$68.00 (KES 8,840)	\$49/mo (KES 6,500/mo)	46 Days
Tier 2: Tactical (Live Demo)	RPi Zero 2 + Sony IMX500 Neural AI Camera + GC9A01 LCD + 4G LTE HAT + 5W Horn	\$148.00 (KES 19,240)	\$73/mo (KES 9,500/mo)	50–68 Days
Tier 3: Solar Pro	30W Solar + LiFePO4 Battery + Sony IMX500 Telephoto + Dual-SIM 4G + 30m IR Array	\$258.00 (KES 33,540)	\$104/mo (KES 13,500/mo)	46–84 Days

3. THE 3 PILLARS OF IBORAIN SAFETY

Pillar 1 — African Transit Fingerprinting™: Reads muddy plates, identifies vehicle customizations, and profiles Boda Bodas (models, helmet compliance, rider jackets, and cargo such as 13kg gas cylinders).

Pillar 2 — Regional Threat Mesh & Instant WhatsApp Alert: Synchronizes hotlists across a 10km grid with sub-600ms latency, triggering active strobes and voice warnings.

Pillar 3 — FreeForm™ Natural Language Crime Search: Detectives query multi-day forensic history using plain English or Swahili (e.g. "Show unplated motorbikes carrying gas cylinders between 1 AM and 4 AM").

4. JOB CREATION & THE CHIPUROBO MICRO-FRANCHISE FLYWHEEL

In direct partnership with **ChipuRobo** (empowering 4M youth by 2030 through hands-on robotics), Iborain trains young technicians from Kenya's informal sector (*Jua Kali*) to assemble, mount, and service sentry nodes. Installers earn **KES 30,400 to KES 58,000/mo (\$233 - \$450/mo)** managing 10 local sentry nodes—creating high-value, sustainable middle-class careers.